Analyzing the Impact of Recession on Automobile Sales

IBM Data Science Capstone Project

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Executive Summary

- Objective
- Analyze the impact of economic recessions on automobile sales.
- Key Findings
- Automobile sales decrease during recession periods.
- Consumer confidence strongly affects sales.
- Unemployment negatively impacts automobile demand.
- Advertising expenditure influences sales performance.
- Interactive dashboards help explore trends effectively

Introduction

- Business Problem
- XYZAutomobiles wants to understand how recessions affect sales performance.
- Goals
- Analyze historical sales trends.
- Identify recession impacts.
- Build visualizations and dashboards.
- Develop predictive models.

Data Collection

- Dataset Features
- Date
- Automobile Sales
- GDP
- Unemployment Rate
- Consumer Confidence
- Vehicle Type
- Advertising Expenditure
- Price
- Recession Indicator

Using the `Ticker` function enter the ticker symbol of the stock we want to extract data on to create a ticker object. The stock is Tesla and its ticker symbol is `TSLA`.

[56]
```
tesla = yf.Ticker("TSLA")
```
Pytho

Using the ticker object and the function `history` extract stock information and save it in a dataframe named `tesla_data`. Set the `period` parameter to `"max"` so we get information for the maximum amount of time.

[57]
```
tesla_data = tesla.history(period="max")
```
Pytho

Reset the index using the `reset_index(inplace=True)` function on the tesla_data DataFrame and display the first five rows of the `tesla_data` dataframe using the `head` function. Take a screenshot of the results and code from the beginning of Question 1 to the results below.

[58]
```
tesla_data.reset_index(inplace=True)
tesla_data.head()
```
Pytho

Data Wrangling

- Steps Performed
- Checked data types.
- Removed unnecessary columns.
- Handled missing values.
- Verified data consistency.
- Prepared data for analysis.

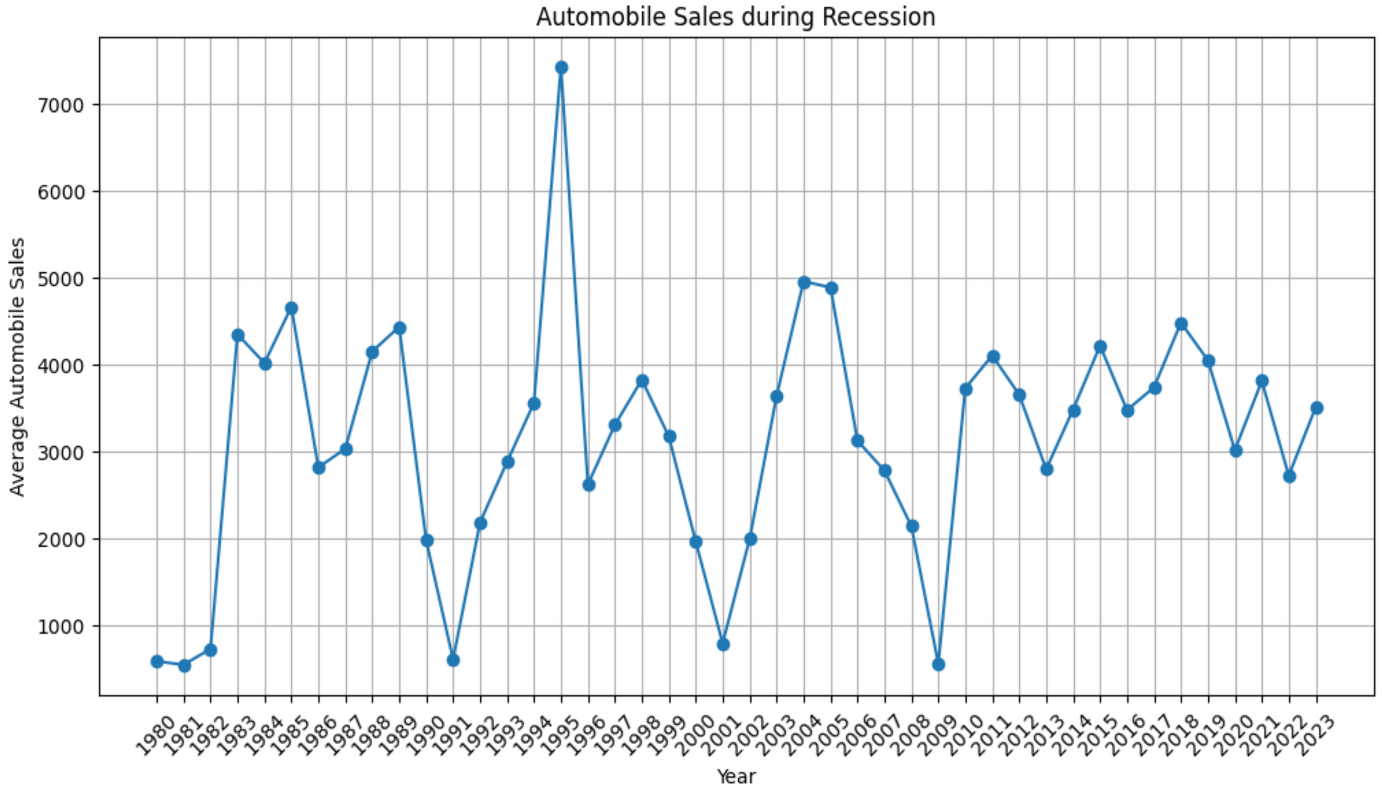
```python
tesla_revenue = pd.DataFrame(columns=["Date", "Revenue"])

for row in soup.find_all("tbody")[1].find_all("tr"):
    col = row.find_all("td")

    if len(col) > 0:
        date = col[0].text
        revenue = col[1].text

        tesla_revenue = pd.concat(
            [tesla_revenue,
             pd.DataFrame({"Date":[date], "Revenue":[revenue]})],
            ignore_index=True
        )
```

EDA Methodology

- Techniques Used
- Line Charts
- Scatter Plots
- Bubble Charts
- Pie Charts
- Seaborn Visualizations
- Folium Maps

Automobile Sales Trends

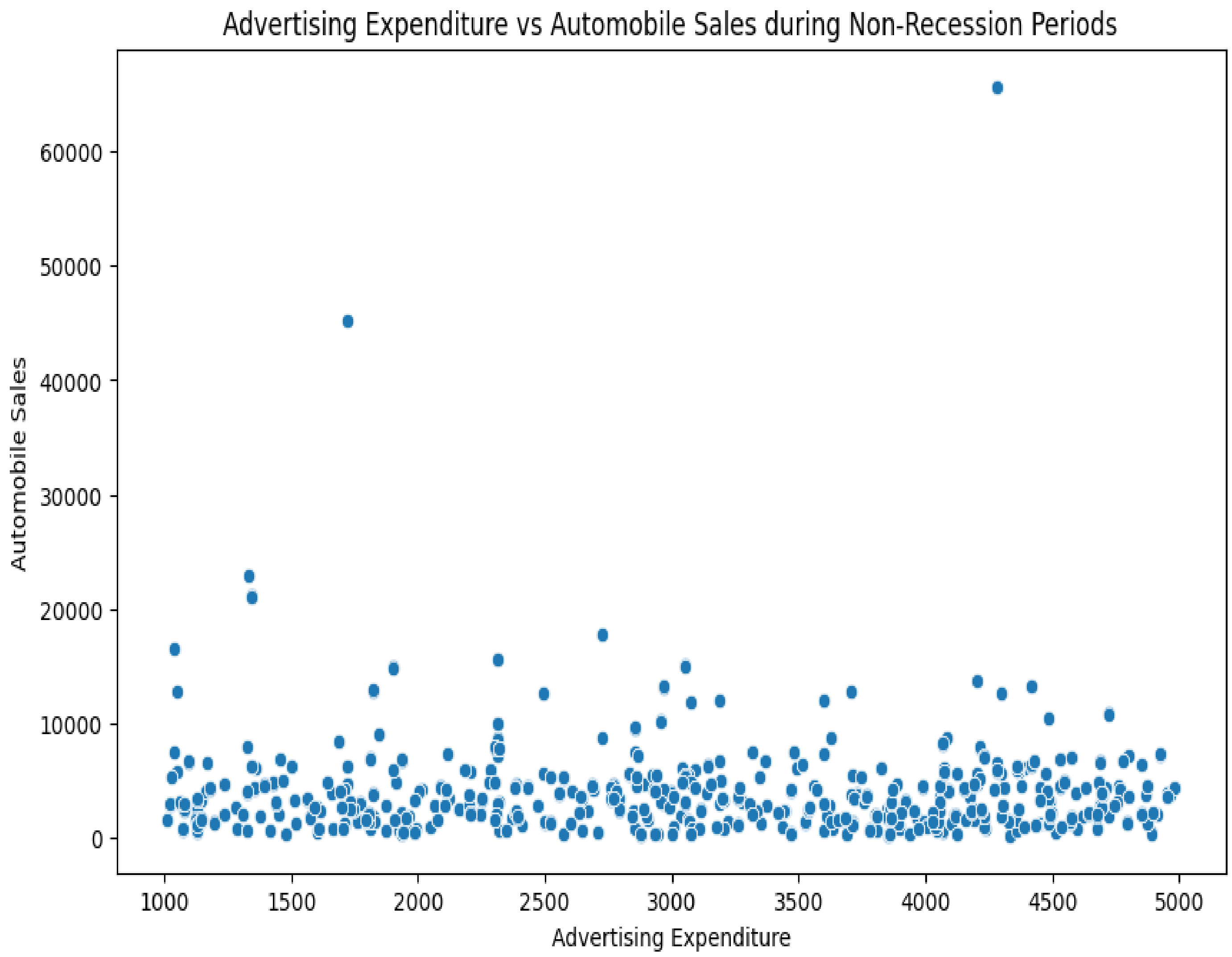
- Insights
- Sales fluctuate yearly.
- Major declines observed during recession periods.
- Recovery occurs after recession ends.



Advertising vs Sales

Insights

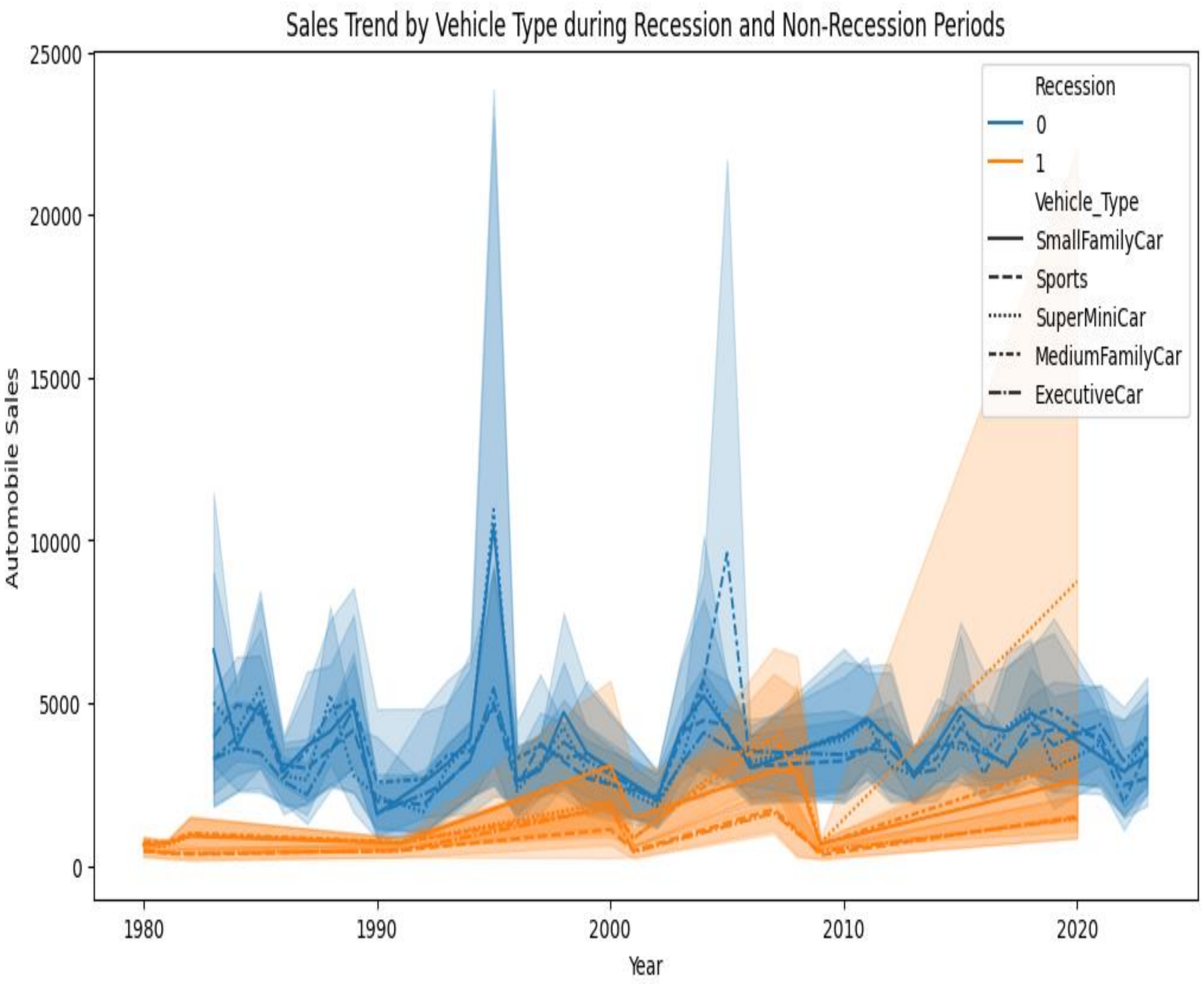
- Positive relationship between advertising expenditure and sales.
- Increased advertising generally leads to higher sales during stable economic periods.



Vehicle Type Analysis

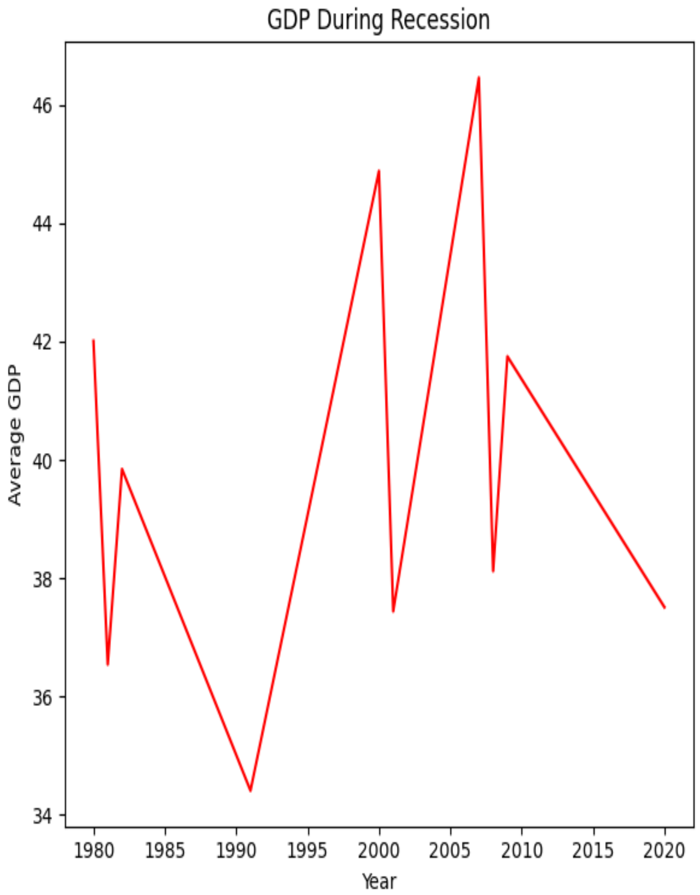
Insights

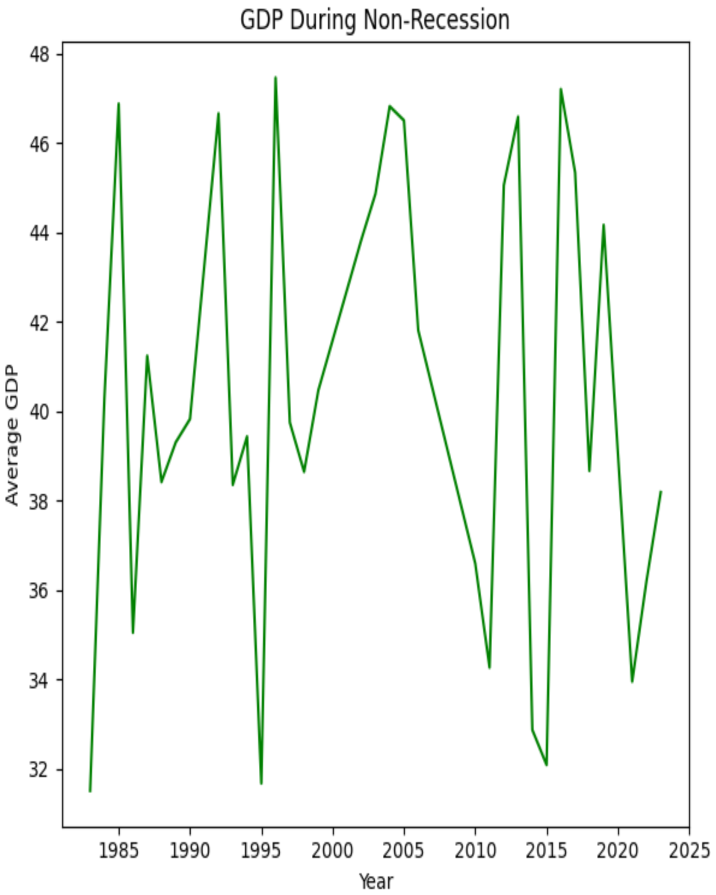
- Small family vehicles maintain stronger sales.
- Sports and luxury vehicles experience larger declines during recessions.



GDP, Seasonality & Consumer Analysis

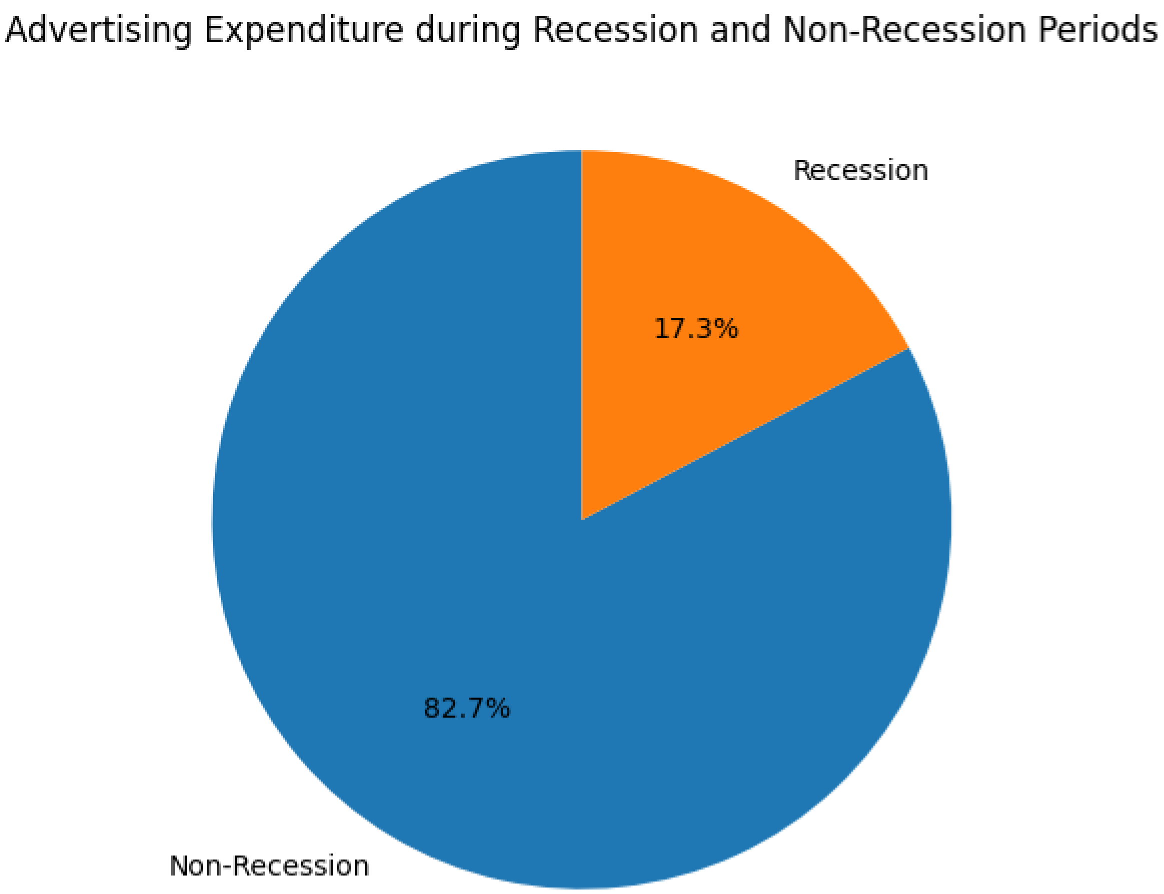
- Findings
- GDP decreases during recessions.
- Seasonality affects automobile sales.
- Consumer confidence positively influences sales.





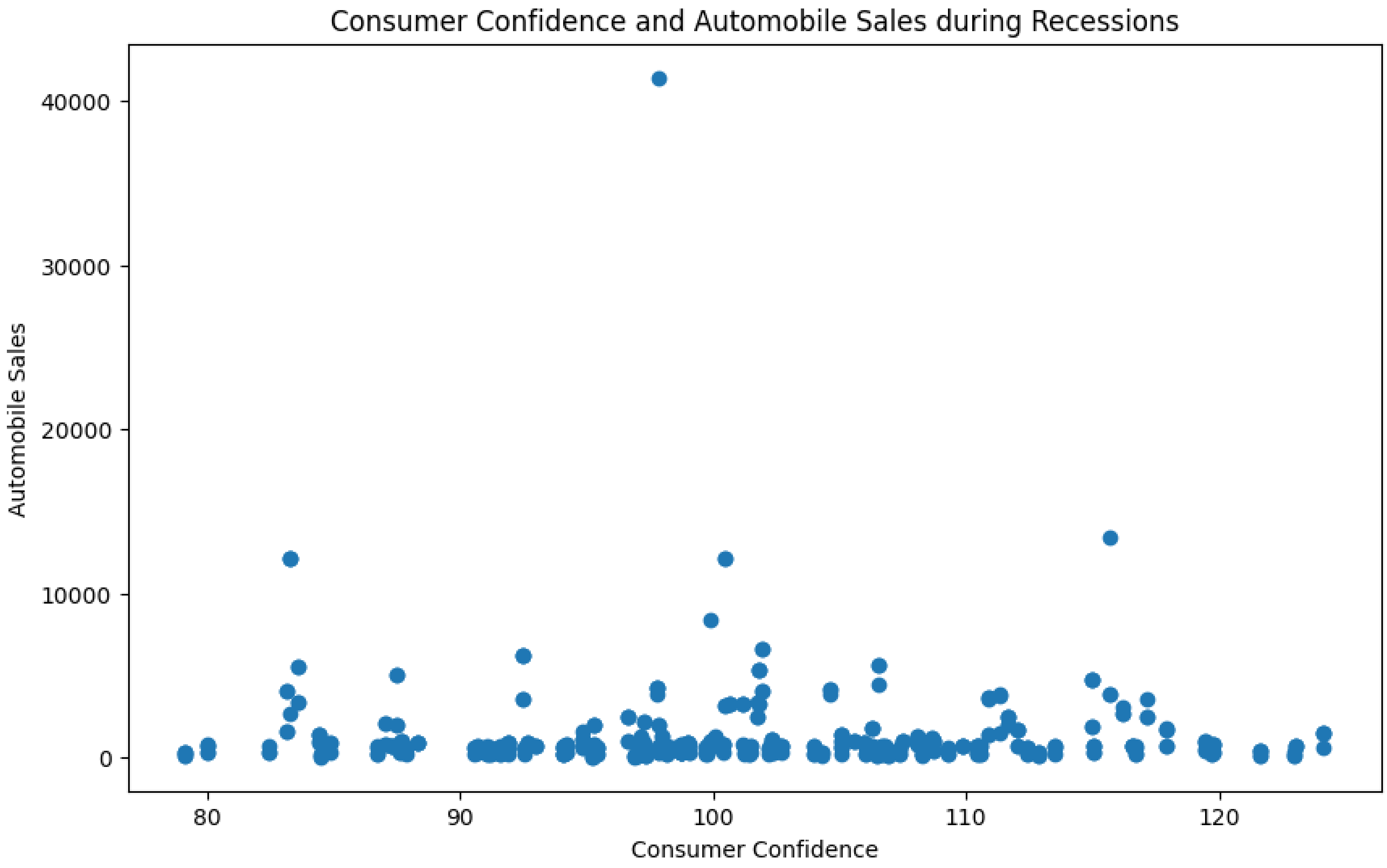
Advertising Expenditure Analysis

- Findings
- Advertising expenditure changes during recession periods.
- Vehicle categories receive different advertising budgets.



Consumer Confidence vs Automobile Sales During Recession

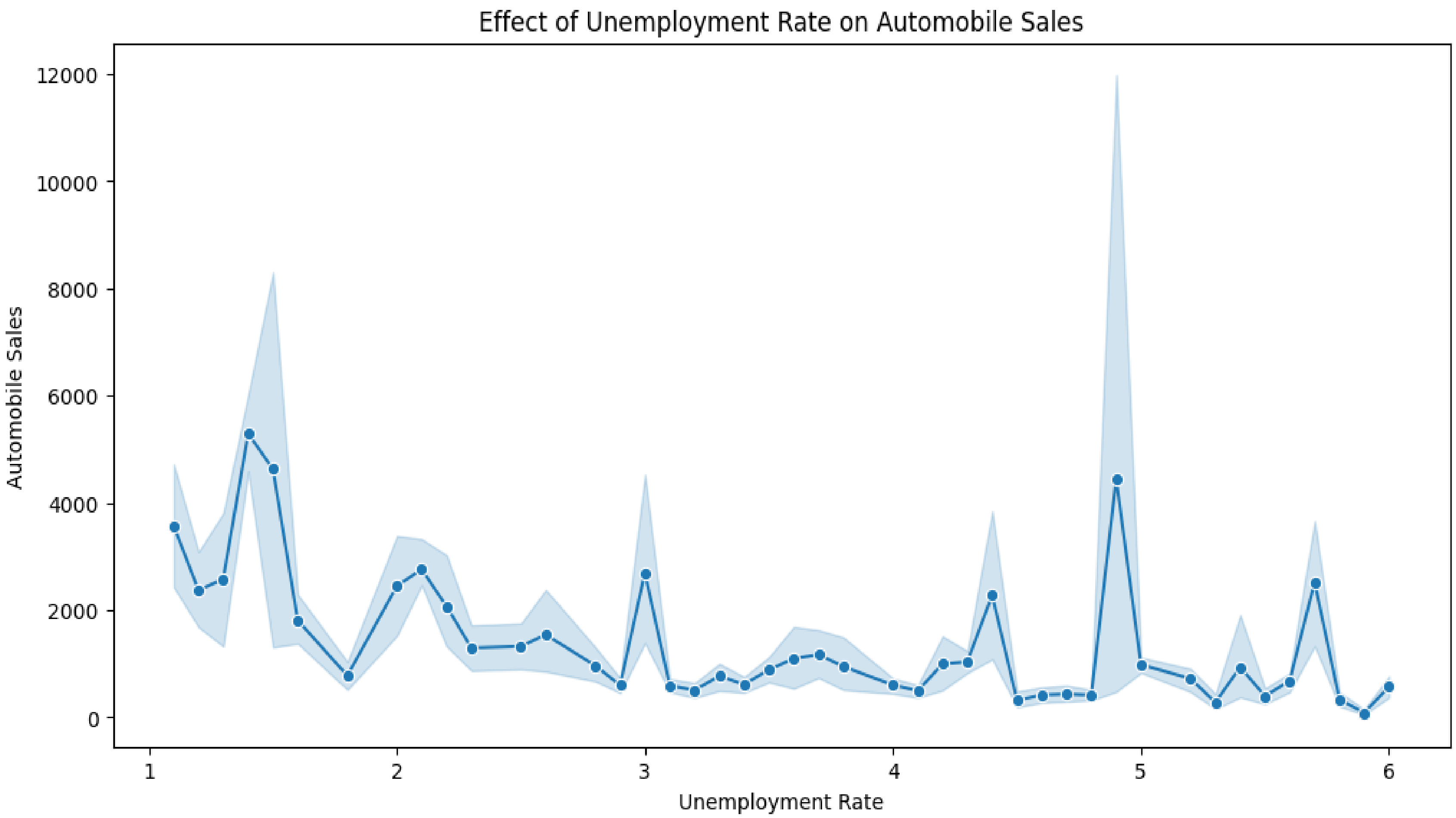
- **Objective**
- Analyze the relationship between consumer confidence and automobile sales during recession periods.
- **Key Findings**
- Consumer confidence has a positive relationship with automobile sales.
- Higher consumer confidence generally corresponds to higher sales volumes.
- During recessions, declining confidence reduces consumer spending on automobiles.
- Consumer sentiment is an important indicator for forecasting vehicle demand.
- **Business Insight**
- XYZAutomobiles should closely monitor consumer confidence levels during economic downturns.
- Marketing campaigns and promotional offers may help maintain sales when confidence declines.


Consumer Confidence and Automobile Sales during Recessions
Automobile Sales
Consumer Confidence
0
10000
20000
30000
40000
80
90
100
110
120

Unemployment & Sales Analysis

Findings

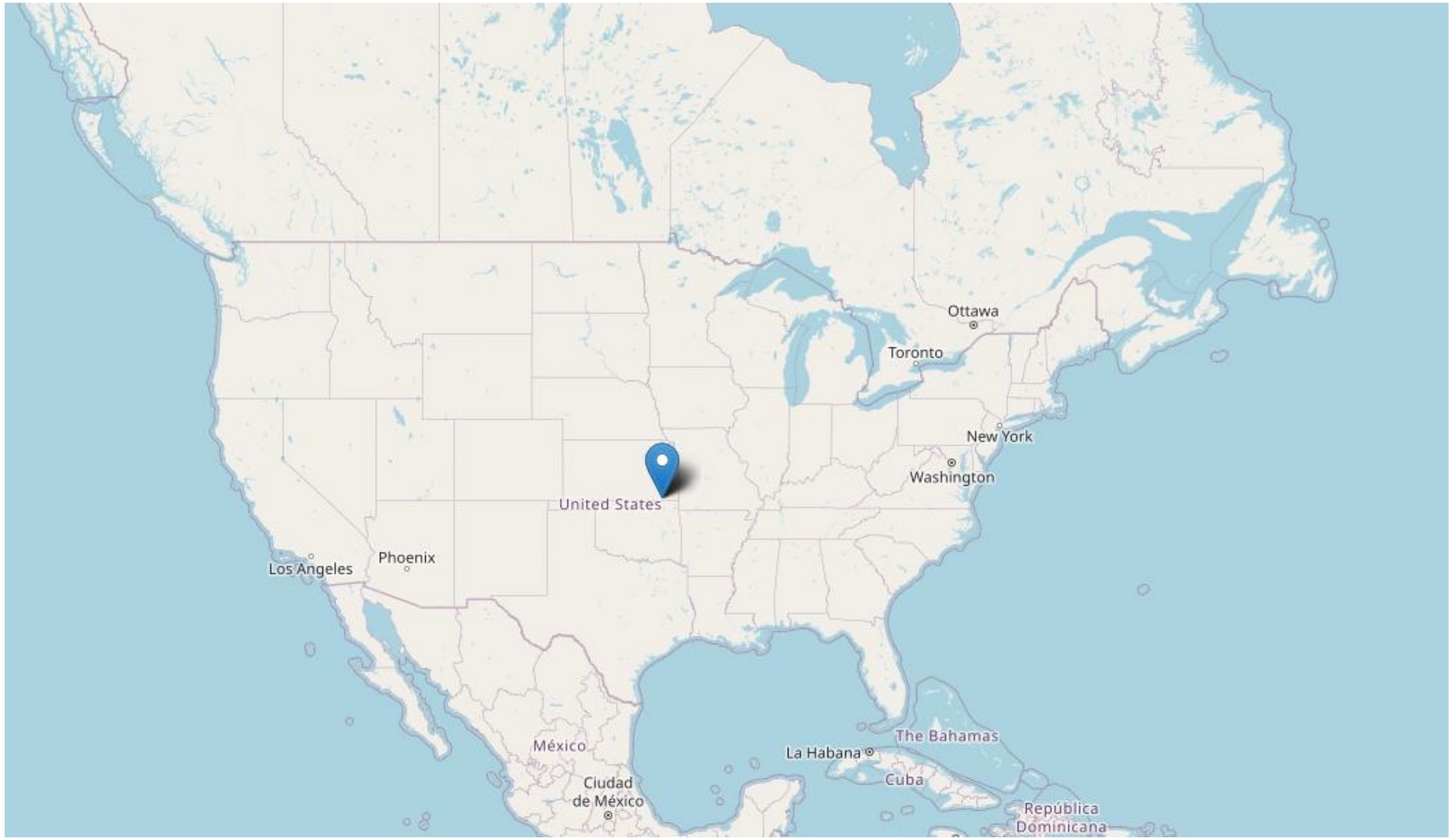
- Higher unemployment rates reduce automobile sales.
- Budget-friendly vehicle types remain more resilient.



Folium Map Results

Findings

- Regional sales patterns identified.
- Certain regions performed better during recession periods.



SQL Analysis Results

- Findings
- SQL queries helped identify trends and patterns.
- Aggregated sales metrics supported business insights.

Plotly Dash Dashboard

- Dashboard Features
- Recession Statistics
- Yearly Statistics
- Interactive Filters
- Dynamic Visualizations

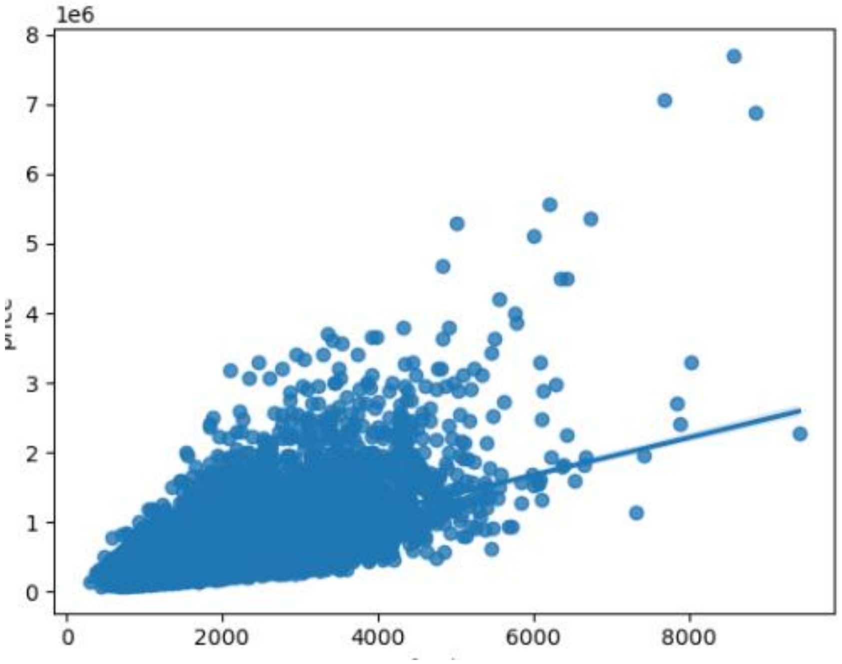
Predictive Analysis Methodology

Models Used

- Linear Regression
- Ridge Regression
- Polynomial Regression

Evaluation Metrics

- R^2 Score
- Mean Squared Error (MSE)



Predictive Analysis Results

Metric	Value
Training R^2	0.91
Test R^2	0.68
MSE	28,500

Results

Interpretation

- Good predictive capability.
- Some overfitting exists.
- Further optimization possible.

Innovative Insights

- Additional Observations
- Consumer confidence is a strong sales indicator.
- Small family cars perform better during economic downturns.
- Advertising remains important even during recessions.

GitHub Repository

- Repository Link
- https://github.com/YOUR_USERNAME/IBM-Data-Science-Capstone
- Include:
- Notebooks
- Python files
- Dashboard code

Conclusion

- Recessions significantly affect automobile sales.
- Consumer confidence and unemployment are major influencing factors.
- Interactive dashboards improve business decision-making.
- Predictive models can support future sales forecasting.

Thank You